

Domain-Complete Evaluation of Distribution Shift in Histopathology

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Paper under double-blind review

Abstract

When a pathology AI model trained at one hospital is deployed at another, it often fails—not because it lacks capacity, but because each hospital’s scanner, staining protocol and patient population leave visual fingerprints that the model confuses with disease. We investigate what actually governs cross-domain generalization by testing eleven robustness methods across three cohorts that isolate different sources of shift: Camelyon17-WILDS (five hospitals, where scanner, stain and patients all change), MIDOG 2021 (three scanners in one laboratory, where only the device changes), and a multi-scanner canine cohort (five scanners, same slides). The central finding is that the *type* of domain shift—not just its presence—determines which method works. Under full hospital shift, self-supervised pre-training on unlabelled tissue lifts the worst hospital’s AUROC by 0.175 and compresses the across-hospital spread threefold; under scanner-only shift, stain normalization dominates instead and pre-training adds nothing. Most domain-generalization objectives (GroupDRO, IRM, DeepCORAL) fail the worst domain on every cohort. A frozen ImageNet linear probe outperforms all trained-from-scratch arms under hospital shift, indicating that representation quality is a stronger lever than any robustness objective at this model scale.

To make these comparisons rigorous, we propose using the intraclass correlation coefficient (ICC_{site}) from a variance decomposition of off-site performance as a quantitative measure of how much of a model’s variability is truly site-driven versus noise from random initialization. On the baseline, hospital identity accounts for 55% of variance; on the best arm it accounts for 84%—meaning the intervention removed the noise floor and left only systematic site structure. We derive the resulting power formula and show that detecting a 0.10 AUROC improvement requires at least eight hospitals at three seeds, far more than current benchmarks provide.

For teams developing pathology AI for clinical deployment, the practical implication is that robustness evaluation must be *domain-complete* (test at every available site), *shift-aware* (know whether you face a scanner change, a protocol change, or a full institutional change), and *seed-replicated* (run multiple seeds to separate real effects from noise). We release the full protocol, all 388 experimental records, and the code to reproduce every result.

1 Introduction

A model trained on slides from one set of hospitals tends to fail at a hospital it has not seen. Each hospital’s scanner, staining chemistry and patient mix leave fingerprints on the tissue image, and a classifier is free to learn those fingerprints whenever they happen to correlate with the diagnosis (Zech et al., 2018; Geirhos et al., 2020). In histopathology the hospital signal is so strong that hospital identity alone is predictable from slide appearance with near-perfect accuracy (Howard et al., 2021), and the practical consequence is well documented: most models that undergo external validation lose accuracy (Yu et al., 2022).

Many methods have been proposed to fix this—from domain-invariant training objectives to stain normalization to self-supervised pre-training—but the way they are evaluated has not kept up. A typical paper tests its fix at one held-out hospital, because that is what the standard benchmark provides: the official

Camelyon17-WILDS split (Koh et al., 2021) designates a single centre as the out-of-distribution test set, and the leaderboard ranks methods by their accuracy there. If your model happens to do well at that particular hospital, the method looks good. If the hospital is an easy one, every method looks good.

This paper asks a simple question: *what happens when you test at every hospital instead of just one?* We hold out each of the five Camelyon17 centres in turn, run eleven robustness interventions through one shared training pipeline with three random seeds per core arm, and then repeat the whole experiment on two additional cohorts where only the scanner changes. The backbone is deliberately small (0.58 M parameters at 64×64 resolution) so that the complete grid—180 GPU jobs—finishes in 32 minutes on one GPU. Any laboratory that can afford a single ablation study can afford this one.

Contributions.

1. **Shift-type-dependent method selection** (§5.1, §5.5). We show that the dominant source of domain shift—scanner hardware, staining protocol, or full institutional change—determines which robustness method works. Under hospital shift (scanner + stain + patients), self-supervised pre-training lifts the worst hospital’s AUROC by 0.175 and compresses cross-hospital spread $3.2\times$; under scanner-only shift, stain normalization dominates instead, and pre-training adds nothing. Most domain-generalization objectives (GroupDRO, IRM, DeepCORAL) hurt the worst domain on every cohort. A frozen ImageNet probe outperforms all trained-from-scratch arms under hospital shift, establishing representation quality as the primary lever at this model scale (§5.2, §5.4).
2. **A variance-decomposition framework for evaluating robustness claims** (§5.6). We propose using the intraclass correlation coefficient ICC_{site} , estimated from a two-way random-effects model, to quantify how much of a model’s performance variability is genuinely site-driven versus initialization noise. We derive a closed-form power formula showing that detecting a 0.10 AUROC mean effect requires at least eight hospitals at three seeds—far beyond what standard single-site evaluations provide. This gives the field a principled tool for deciding when a robustness claim is statistically supported.
3. **A domain-complete evaluation protocol for clinical AI deployment** (§3). We operationalize these findings into a concrete protocol: evaluate at every available domain, report worst-domain performance and across-domain spread, replicate seeds, and use the ICC decomposition to verify that the evaluation grid has sufficient power before publishing a robustness claim. At compact-model scale the complete protocol costs under one hour of GPU time.

All 388 experimental records and the code to regenerate every table and figure are publicly released.

2 Related work

Site effects in medical imaging. That acquisition site drives model behaviour is established across modalities (Zech et al., 2018; Yu et al., 2022). Howard et al. (2021) showed for histopathology specifically that site signatures are recoverable from slide appearance and that they bias downstream predictions. Our contribution is not the existence of the effect but its magnitude relative to method effects when both are measured under one protocol.

Domain generalization and the WILDS benchmark. Camelyon17-WILDS (Koh et al., 2021; Bandi et al., 2019) is the standard testbed for cross-hospital shift. Methods evaluated on it include group distributionally robust optimization (Sagawa et al., 2020), invariant risk minimization (Arjovsky et al., 2019), deep feature-covariance alignment (Sun & Saenko, 2016), style mixing (Zhou et al., 2021), gradient matching (Shi et al., 2022) and selective augmentation (Yao et al., 2022). Each is reported at the benchmark’s single out-of-distribution test centre. We re-implement six of them inside our pipeline and run each at all five centres.

Self-supervision and augmentation in histopathology. Contrastive pre-training on unlabelled tissue is an established recipe in computational pathology (Ciga et al., 2022), as is aggressive stain-space augmentation (Tellez et al., 2019) and classical stain normalization (Macenko et al., 2009). We make no claim of methodological novelty for the SimCLR arm (Chen et al., 2020); the claim is about what a site-complete evaluation reveals about it relative to the alternatives. Pathology foundation models (Chen et al., 2024) operate two or more orders of magnitude above our scale; §6 discusses what that means for our conclusions.

Variance in benchmarks. Bouthillier et al. (2021) argued that machine-learning benchmarks under-report sources of variance and that conclusions often do not survive accounting for them. Our variance decomposition is an instance of that programme for one concrete clinical benchmark, with the twist that one of the variance sources, hospital identity, is also the quantity of scientific interest.

3 The site-complete evaluation

Protocol. Camelyon17-WILDS supplies 96×96 lymph-node patches from five hospitals, labelled for tumour in the central 32×32 region; we centre-crop to 64×64 . For each fold $t \in \{0, \dots, 4\}$ we hold out centre t as the off-site test hospital, use centre $v = (t - 1) \bmod 5$ as the out-of-distribution validation hospital for epoch selection, and train on the remaining three centres. Each fold draws 30,000 class-stratified training patches, up to 12,000 in-distribution validation patches, 12,000 out-of-distribution validation patches and 25,000 off-site test patches, matching the official WILDS budget; 40,000 further patches from the training centres are held out unlabelled for pre-training. Every fold is verified patient-disjoint across all roles, and each fold’s self-supervised encoder sees only that fold’s own unlabelled pool, so no model is ever pre-trained on tissue from its own test hospital. Fold $t = 4$ (train $\{0, 1, 2\}$, validate on 3, test on 4) reproduces the official WILDS Camelyon17 split exactly, which lets us locate the benchmark’s canonical configuration inside the grid.

What to report. A K -hospital grid supports some summaries better than others. We report three. The *worst-site* value is the AUROC at the hospital where the arm does worst; it is the statistic a deployment decision actually consumes, and it is well defined for any K . The *across-site standard deviation* measures how much the choice of hospital still matters once the intervention is applied. The *mean across sites* is the conventional summary, and we report it with a paired-across-hospitals confidence interval together with the number of hospitals that interval would need to be informative at a given effect size (§5.6).

4 Experimental setup

All non-probe arms share one backbone: four convolutional stages of width 32/64/128/256 with batch normalization, global average pooling and a linear classifier, 583,394 parameters in total. All share one fine-tuning schedule: SGD with momentum 0.9, cosine learning rate from 0.02, weight decay 10^{-4} , batch size 128, five epochs, with the epoch chosen by out-of-distribution validation AUROC. No per-arm hyperparameter tuning was performed; every method uses its authors’ recommended defaults. The arms are:

- **ERM** — the supervised baseline.
- **SAM** — sharpness-aware minimization (Foret et al., 2021) at $\rho = 0.05$, batch-norm statistics frozen on the ascent step.
- **SSL** (ours) — a SimCLR encoder (Chen et al., 2020) pre-trained on the fold’s own unlabelled pool with NT-Xent loss at temperature 0.5, batch size 256, six epochs, H&E-style colour and stain jitter, followed by the shared supervised fine-tune.
- **SSL+SAM** (ours) — the same encoder, fine-tuned with SAM.
- Four domain-generalization objectives: **GroupDRO** (Sagawa et al., 2020) at $\eta = 0.01$, **DeepCORAL** (Sun & Saenko, 2016) at $\lambda = 1$, **IRMv1** (Arjovsky et al., 2019) at $\lambda = 1$ annealed from epoch

2, and **MixStyle** (Zhou et al., 2021) at $p = 0.5, \alpha = 0.1$ after stage 2. Hospital identity supplies the group/domain label.

- **Macenko + ERM** — stain normalization (Macenko et al., 2009) fitted on a 64-tile reference sample from the training set and applied to every input.
- Three methods from the WILDS leaderboard: **Fish** (Shi et al., 2022) with meta-learning rate 0.1 and hospital-grouped inner steps, **LISA** (Yao et al., 2022) with same-class inter-hospital Beta(2, 2) mix-up, and **ERM + H&E jitter**, aggressive colour/brightness/hue jitter in H&E space at strength 0.4, following the augmentation-only robustness strategy of Tellez et al. (2019) and in the spirit of the strong-augmentation baselines of Taori et al. (2020).
- **Frozen probe** — a linear head on the frozen 2048-d penultimate features of a torchvision ResNet50 (He et al., 2016) pre-trained on ImageNet-1K (Deng et al., 2009), inputs bilinearly upsampled to 224×224 , head trained 15 epochs under the same selection rule.
- Two further arms, used on the scanner cohorts of §5.5, reproduce the published top performers on those data: **TIA-style**, the stain normalization plus heavy colour and dihedral augmentation of the classification stage of the MIDOG-winning pipeline (Jahanifar et al., 2024), and **RotInv**, the dihedral invariance and hard-negative strategy of Lafarge & Koelzer (2021).

The four core arms (ERM, SAM, SSL, SSL+SAM) are run at three seeds per fold; the remaining seven at one seed per fold. We report AUROC, average precision, accuracy, F1 and expected calibration error over 15 equal-width bins (Naeini et al., 2015; Guo et al., 2017). The Camelyon17 grid comes to 15 self-supervised pre-training jobs, 105 supervised training runs and 60 PatchCamelyon checkpoint evaluations: 180 GPU jobs, 0.53 H100-hours. With the two scanner cohorts of §5.5 the study totals 388 GPU jobs and 1.23 H100-hours.

5 Results

The results tell a story in three parts. We first show the full picture of which methods survive testing at all five hospitals and which do not (§5.1), then look closely at why self-supervised pre-training works where others fail (§5.2–5.4), and finally reveal that the entire ranking flips when we change what kind of shift the model faces (§5.5). We close with the practical design question of how many hospitals and random seeds a reliable evaluation actually requires (§5.6).

5.1 Most interventions fail domain-complete evaluation

The first thing the domain-complete grid reveals is how much the test hospital matters. The supervised baseline’s off-site AUROC runs from 0.724 at centre 2 to 0.920 at centre 0—a spread of 0.196 produced by the choice of test hospital alone (Figure 1a). Every method contrast in the study is smaller than that spread. A report at centre 0 would describe the baseline as strong; a report at centre 2 would describe the same pipeline on the same data as barely usable. The official WILDS split (fold 4) sits at 0.784, neither the best nor the worst draw.

Table 1 places all thirteen arms on the same grid, ordered by worst-site AUROC; Figure 1b plots the two primary summaries against each other. Three patterns emerge.

First, only four of the ten alternatives improve the worst hospital at all: the frozen probe (0.929), H&E-space input jitter (0.871), MixStyle (0.772), and SAM by 0.002 AUROC, which is noise. The other six leave the worst hospital *below* the supervised baseline, among them three of the four domain-generalization objectives, classical stain normalization, and both leaderboard methods other than H&E jitter. Under this protocol the two families that work are representation-space pre-training and input-space augmentation, and they work to comparable degrees. ERM with H&E jitter reaches worst-site 0.871 and mean 0.944, close to SSL+SAM, with no pre-training stage at all. That favours the simpler method, and we report it as such.

Second, the spread the single-centre convention hides can be very large. Fish ranges from 0.533 to 0.919 AUROC across the five hospitals of one cohort, and LISA from 0.505 to 0.906. A number reported at one

Table 1: Thirteen arms under the site-complete evaluation, ordered by worst-site AUROC. Seed SD is the mean over folds of the within-fold seed standard deviation, available only for the multi-seed arms. r is the correlation between per-hospital gain over ERM and per-hospital ERM AUROC; r_0 is its value under independence (Appendix F); the informative quantity is the excess $r - r_0$. ECE is averaged over hospitals.

Arm	Worst site	Across-site SD	Mean	Seed SD	ECE	r	r_0	$r - r_0$
Frozen probe (IN-1K)	0.929	0.019	0.949	—	0.022	−0.98	−0.98	−0.00
SSL (ours)	0.899	0.026	0.933	0.021	0.073	−0.95	−0.96	+0.00
SSL+SAM (ours)	0.894	0.028	0.941	0.009	0.045	−0.96	−0.95	−0.01
ERM + H&E jitter	0.871	0.042	0.944	—	0.041	−0.89	−0.89	+0.00
MixStyle	0.772	0.070	0.872	—	0.202	−0.56	−0.77	+0.21
SAM	0.726	0.074	0.828	0.084	0.185	−0.69	−0.75	+0.06
ERM baseline	0.724	0.083	0.838	0.054	0.187	—	—	—
GroupDRO	0.690	0.095	0.813	—	0.188	+0.71	−0.66	+1.37
DeepCORAL	0.683	0.101	0.814	—	0.179	−0.12	−0.64	+0.51
IRMv1	0.673	0.103	0.772	—	0.229	−0.28	−0.63	+0.35
Macenko + ERM	0.660	0.107	0.825	—	0.195	−0.15	−0.62	+0.46
Fish	0.533	0.205	0.689	—	0.273	+0.60	−0.38	+0.97
LISA	0.505	0.162	0.761	—	0.281	−0.54	−0.46	−0.09

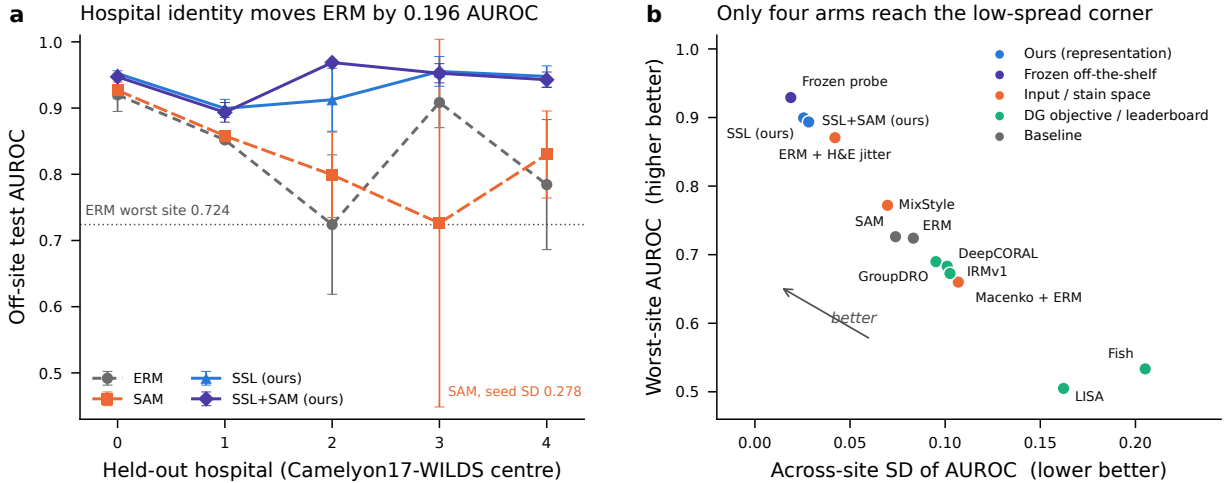


Figure 1: (a) Off-site AUROC at each held-out hospital, mean over three seeds, error bars one seed standard deviation. The supervised baseline swings 0.196 AUROC across hospitals; the pre-trained arms are nearly flat. SAM’s error bar at centre 3 spans 0.278 AUROC, so seed noise alone, at one hospital, exceeds the entire site range of the baseline. (b) All thirteen arms in the worst-site / across-site-spread plane. Only the frozen probe, the two pre-trained arms and H&E jitter reach the low-spread corner.

centre for either method describes that centre. It is worth being careful about what this does and does not show. Our re-implementations run inside a 0.58 M-parameter, five-epoch pipeline rather than the authors’ own, and Fish’s out-of-distribution validation curve selects epoch 1 in four of five folds (Figure 10), which is as much a model-selection pathology as a method failure. What we are claiming is that these methods are highly sensitive to which hospital is held out, not that these are the best numbers they can reach.

Third, a linear probe on a frozen ImageNet-1K ResNet50 beats every trained-from-scratch arm on every statistic in the table: worst-site 0.929, across-site SD 0.019, mean 0.949, ECE 0.022. It does so after 22 seconds of head-only training per fold, with no augmentation and no pathology-specific supervision. This is not a pathology-foundation-model result and we do not present it as one. It is a reference point: one step up in representation quality leaves the hospital-to-hospital range nearly flat, which is what the site-shortcut

Table 2: The core grid: off-site test AUROC at each held-out hospital, mean over three seeds $\pm$ the within-cell seed standard deviation. δ columns are differences from the ERM cell of the same hospital; $I = \delta_{\text{SSL}+\text{SAM}} - \delta_{\text{SSL}} - \delta_{\text{SAM}}$ is the interaction. Fold 4 is the official WILDS split.

Held-out	ERM	SAM	SSL	SSL+SAM	δ_{SSL}	δ_{SAM}	I
centre 0	0.920 $\pm$.025	0.927 $\pm$.012	0.952 $\pm$.004	0.947 $\pm$.002	+0.032	+0.007	-0.012
centre 1	0.852 $\pm$.003	0.858 $\pm$.001	0.899 $\pm$.014	0.894 $\pm$.015	+0.047	+0.006	-0.012
centre 2	0.724 $\pm$.105	0.799 $\pm$.065	0.913 $\pm$.048	0.969 $\pm$.002	+0.189	+0.075	-0.019
centre 3	0.908 $\pm$.038	0.726 $\pm$.278	0.955 $\pm$.022	0.953 $\pm$.014	+0.047	-0.182	+0.180
centre 4	0.784 $\pm$.098	0.830 $\pm$.066	0.948 $\pm$.016	0.943 $\pm$.012	+0.163	+0.046	-0.050
Mean	0.838	0.828	0.933	0.941	+0.096	-0.010	+0.017
Across-site SD	0.083	0.074	0.026	0.028	—	—	—
Worst site	0.724	0.726	0.899	0.894	—	—	—
Range	0.196	0.200	0.056	0.075	—	—	—
Seed SD	0.054	0.084	0.021	0.009	—	—	—

account predicts a sufficiently general representation should do, and it sets a bar that none of the eleven trained-from-scratch interventions clears.

The last three columns of Table 1 address whether the arms that do help target the hospitals that need it most. A large negative correlation r between a method’s per-hospital gain and the baseline’s per-hospital AUROC looks like evidence of such targeting, but the statistic is mechanically negative for any arm that is flatter than the baseline. For per-hospital performance that is statistically independent of the baseline’s,

$$r_0 = \frac{-\sigma_E}{\sqrt{\sigma_M^2 + \sigma_E^2}}, \quad (1)$$

where σ_E and σ_M are the across-site standard deviations of the baseline and method respectively; an arm that is simply uniform ($\sigma_M \rightarrow 0$) has $r_0 \rightarrow -1$ with no targeting. The excess $r - r_0$ removes this artifact. Our pre-trained arms sit at zero excess—they are uniformly better, not selectively repairing the hard hospitals. GroupDRO (+1.37) and Fish (+0.97) show the opposite: they concentrate what gain they have at the hospitals the baseline already handles. The full derivation and per-arm response curves are in Appendix F.

5.2 Self-supervised pre-training compresses both nuisances

Table 2 breaks the four core arms out by hospital with three seeds per cell. The pre-trained arms stand out from the grid above: they behave the way a site-robustness intervention should. SSL raises the worst hospital from 0.724 to 0.899 AUROC (+0.175) and compresses the across-site standard deviation from 0.083 to 0.026, a 3.2 $\times$ reduction; the range across hospitals falls from 0.196 to 0.056. The mean effect is +0.096 AUROC with a paired-across-hospitals 95% confidence interval of [+0.004, +0.188] ($t_4 = 2.89$, $p = 0.045$). That is significant, but by a margin that Table 6 says we should not lean on at $K = 5$, so we lean instead on the worst-site and spread numbers, which are large and consistent.

Sharpness-aware minimization on its own does none of this. Its mean effect is -0.010 AUROC (95% CI [-0.135, +0.115], $p = 0.84$), its sign varies by hospital, and it is the least stable arm in the study. Combined with pre-training, though, it contributes something SSL alone does not: SSL+SAM has the lowest seed standard deviation of any arm at 0.009, a sixfold reduction on the baseline’s 0.054 and less than half SSL’s 0.021. The two mechanisms do not interact in the mean. The interaction term I has a five-hospital mean of +0.017 AUROC (95% CI [-0.097, +0.132], $p = 0.70$), indistinguishable from zero, so the discrimination gain of the combined arm is attributable to pre-training alone. What SAM adds on top of pre-training is reproducibility and calibration, not accuracy.

5.3 Calibration at the unseen hospital

A workflow that thresholds on model confidence inherits the calibration the model has at the point of use, and calibration degrades under shift more sharply than discrimination does (Guo et al., 2017). This is where

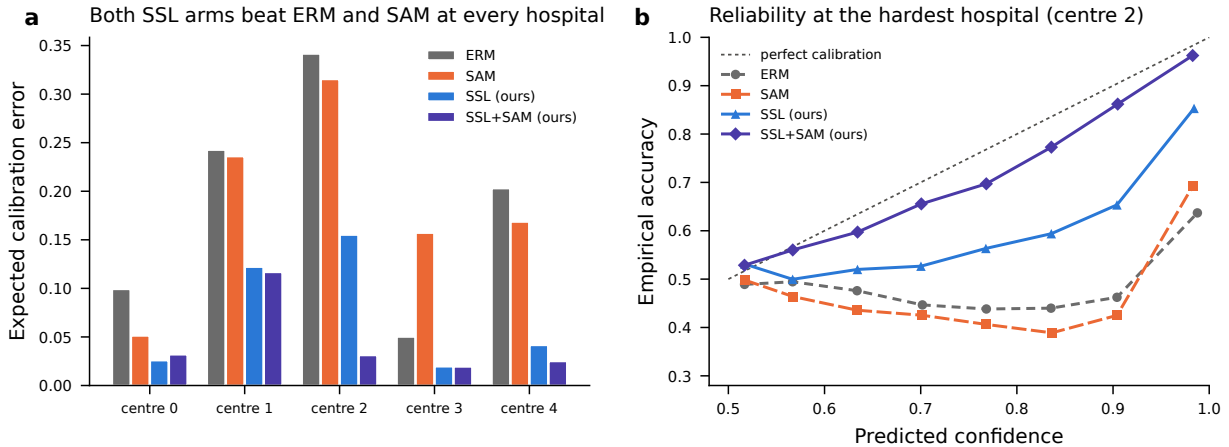


Figure 2: (a) Expected calibration error at each held-out hospital, seed-averaged. (b) Reliability at the hardest hospital (centre 2), three seeds pooled, 15 equal-width bins. ERM and SAM are anti-calibrated over most of the range: higher confidence, lower accuracy.

Table 3: Out-of-partition validation on PatchCamelyon. Statistics are computed across the five WILDS folds after averaging the three seeds within each fold. No PatchCamelyon example was seen in training, model selection or tuning.

Arm	Mean AUROC	Across-fold SD	Worst fold	Best fold	ECE
ERM	0.707	0.095	0.541	0.768	0.294
SAM	0.722	0.092	0.562	0.798	0.300
SSL	0.753	0.046	0.676	0.786	0.258
SSL+SAM	0.779	0.014	0.762	0.797	0.225

the pre-trained arms help most clearly. Averaged over hospitals, expected calibration error falls from 0.187 (ERM) and 0.185 (SAM) to 0.073 (SSL) and 0.045 (SSL+SAM); at the worst hospital it falls from 0.341 to 0.117. Both pre-trained arms beat both non-pre-trained arms at *every* one of the five hospitals (Figure 2a). The reliability diagram at the hardest hospital (Figure 2b) shows something the summary statistic hides. ERM and SAM are not simply mis-scaled; they are non-monotone, growing *less* accurate as they grow more confident across most of the confidence range. That is the failure mode that matters when confidence is used to triage, and it is the one that a single ECE number is least likely to expose. SSL+SAM tracks the diagonal.

5.4 Out-of-partition validation on PatchCamelyon

A site-complete result inside Camelyon17-WILDS could still be a property of that partition. We therefore evaluated all 60 checkpoints of the multi-seed grid (5 folds $\times$ 4 arms $\times$ 3 seeds) on the 32,768-patch PatchCamelyon test split (Veeling et al., 2018), a different partition of the Camelyon cohort, with no further training, model selection or tuning. Table 3 and Figure 3 give the result.

The absolute AUROC drops sharply, from 0.94 in cohort to 0.78. We take that as a sobering measure of how little a 0.58M-parameter model trained at 64×64 transfers, and we return to it in §6. The structure of the result nonetheless reproduces. The across-fold standard deviation falls from 0.095 (ERM) to 0.014 (SSL+SAM), a $6.8\times$ compression that is *larger* than the $2.9\times$ seen within WILDS, and the worst fold rises by 0.221 AUROC, again larger than the in-cohort floor lift of 0.169. The calibration ordering is preserved too. Whatever pre-training is doing, it is a property of the learned representation and not of the WILDS partition.

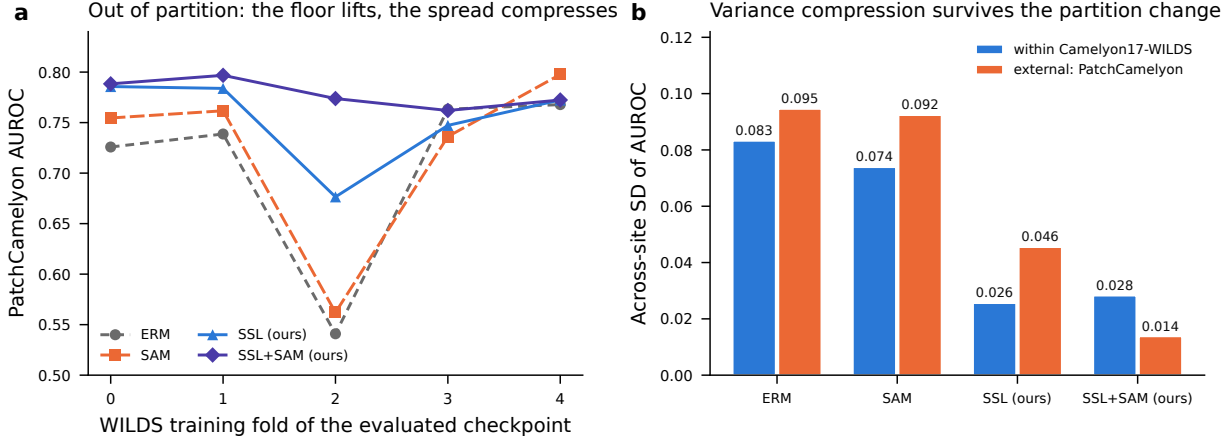


Figure 3: (a) PatchCamelyon AUROC of every checkpoint group, by the WILDS fold it was trained on. (b) Across-site standard deviation within WILDS and on PatchCamelyon. The compression relative to ERM survives the partition change and strengthens for SSL+SAM.

5.5 The method ranking depends on the kind of shift

Everything so far rests on one cohort, and Camelyon17’s site variable bundles three things together: the scanner, the staining and the patient population. A natural question is which of the three the effect comes from, and whether any of the conclusions above are properties of Camelyon17 rather than of cross-domain evaluation. We therefore repeated the whole grid on two cohorts where the scanner is the *only* thing that varies.

MIDOG 2021 (Aubreville et al., 2023) is 200 breast-cancer regions, 50 from each of four scanners, every one of them prepared in the same laboratory. Three scanners carry annotations, which are binary: mitotic figure versus a hard non-mitotic look-alike. Discriminating those two is the second stage of the pipeline that won both MIDOG 2021 and 2022 (Jahanifar et al., 2024), so the task sits inside the published problem. Holding the lab constant removes the staining and preparation component of site and leaves the acquisition device. $K = 3$ labelled scanners is one short of what the Camelyon17 protocol needs, so there is no spare domain to validate on and epochs are selected on a held-out split of the two training scanners; and because MIDOG carries about 2.8k training patches per fold rather than 30k, we match total optimizer steps (50 epochs) instead of epochs. Everything else is unchanged.

Canine SCC (Wilm et al., 2023) is 44 canine cutaneous squamous cell carcinoma slides, each digitized on five scanners, with tumour and six tissue classes annotated once and transferred to the other four scanners by registration. We classify tumour against non-tumour tissue. This cohort is the one place where the Camelyon17 protocol transfers with no adaptation at all: $K = 5$ domains, a genuine held-out validation scanner, and 24.5k–28.6k training patches per fold, so the same five-epoch schedule applies. Two details matter. The scanners differ in resolution, so a fixed pixel crop would make part of the “domain effect” a trivial zoom; because the annotations are registered, the total annotated area per slide gives a precise scale ratio (coefficient of variation 0.001 across slides, matching the scanners’ published $\mu\text{m}/\text{px}$), and we crop each scanner at its own pixel size and resize, so patches cover a constant physical extent. And since the same 44 slides appear under all five scanners, we also partition slides into five disjoint groups, so a held-out fold withholds a scanner *and* a set of slides, as a held-out hospital in Camelyon17 withholds patients.

We also added the two published MIDOG top performers as arms, which brings the grid to fifteen: **TIA-style**, the stain normalization plus heavy colour and orientation augmentation of the challenge-winning classification stage (Jahanifar et al., 2024), and **RotInv**, the dihedral-invariance and hard-negative strategy of Lafarge & Koelzer (2021).

Table 4 and Figures 4 and 5 give the result, and it splits cleanly in two.

Table 4: The same grid on three cohorts, all through one pipeline and one aggregator. σ_{site} and σ_{run} are the ERM variance components; $K_{0.10}$ is the number of domains needed to detect a 0.10 AUROC mean effect at three seeds. Full per-arm tables are in Appendix C.

Cohort	Domain is	K	σ_{site}	σ_{run}	ICC	Best arm (worst-site)	SSL effect	$K_{0.10}$
Camelyon17	hospital	5	0.074	0.068	0.55	Frozen probe 0.929	+0.096*	8
MIDOG 2021	scanner, one lab	3	0.025	0.039	0.30	Macenko 0.865	+0.028	4
Canine SCC	scanner, 5 devices	5	0.050	0.040	0.61	Frozen probe 0.792	+0.012	5

*the only cohort where the SSL mean effect is significant at $\alpha = 0.05$ ($p = 0.045$; MIDOG $p = 0.17$, canine $p = 0.43$).

The design conclusions replicate; the method ranking does not. Both design claims hold on all three cohorts. Every cohort needs more than one domain— $K_{0.10}$ is 8, 4 and 5 at three seeds, never 1—and on every cohort a sizeable share of the variance is not site: the intraclass correlation is 0.55, 0.30 and 0.61, and on MIDOG the run-to-run term $\sigma_{\text{run}} = 0.039$ actually *exceeds* the between-scanner term $\sigma_{\text{site}} = 0.025$. Fish and GroupDRO sit near the bottom of all three tables. But the magnitude of the domain effect tracks what varies: ERM’s between-domain SD falls from 0.074 when the domain is a hospital to 0.050 and 0.025 when it is only a scanner (Figure 4a). On this evidence the scanner accounts for perhaps a third to two thirds of the Camelyon17 hospital effect, and stain, preparation and case mix account for the rest.

Stain normalization and the frozen probe swap places. Macenko stain normalization is eleventh of thirteen on Camelyon17 and *first of fifteen* on MIDOG, where it reaches worst-scanner AUROC 0.865 against ERM’s 0.754. The frozen ImageNet probe is first on Camelyon17 and first on canine but ninth on MIDOG. Figure 5a plots the whole reordering. Both reversals make mechanistic sense once stated: MIDOG’s shift is purely one device’s colour rendition of the same physical slide, which is exactly what stain normalization is built to undo, whereas the frozen ImageNet features that separate tumour from normal tissue so well carry little about the fine nuclear morphology that distinguishes a mitotic figure from its look-alike. The practical consequence is uncomfortable and worth stating plainly: a method’s robustness ranking cannot be read off one benchmark even when that benchmark is evaluated domain-completely, because the *kind* of shift matters as much as which domain is held out.

Our own positive claim narrows. The SSL mean effect is +0.096 AUROC on Camelyon17 ($p = 0.045$), +0.028 on MIDOG ($p = 0.17$) and +0.012 on canine ($p = 0.43$); SSL’s rank by worst domain goes from second to fifth to fourth (Figure 5b). Pre-training on in-domain unlabelled tissue earns its cost when the shift includes a change of laboratory and patient population, and not when the shift is only the acquisition device. That is consistent with the NCT-CRC-HE result in Appendix D, where a same-institution stain perturbation also failed to reward pre-training, and it replaces the broad claim we would have made from Camelyon17 alone with a narrower one we can actually support. What does hold up across both scanner cohorts is the published MIDOG-winning recipe: TIA-style is second by worst-domain AUROC on MIDOG *and* on canine, and the best-calibrated trained-from-scratch arm on both.

5.6 How many domains and seeds does an evaluation need?

It would be tempting to read the results above as evidence that the test hospital dominates everything. The seed-level detail in Table 2 says otherwise. SAM’s seed standard deviation at centre 3 is 0.278 AUROC—reruns of one configuration at one hospital move the result further than the entire across-hospital range of the baseline. Fitting the two-way random-effects model $y_{ts} = \mu + b_t + e_{s(t)}$ to each arm (Searle et al., 1992) separates the two sources (Table 5, Figure 6a).

Hospital identity accounts for 55% of the baseline’s variance and only 4% of SAM’s (Table 5). So single-hospital, single-seed evaluation is under-powered in two directions at once, and averaging more seeds does not substitute for adding hospitals any more than the reverse does. The decomposition is also useful diagnostically. SAM’s across-site standard deviation of 0.074 looks comparable to the baseline’s, but almost none of it is systematic site structure: the spread is instability, not site sensitivity.

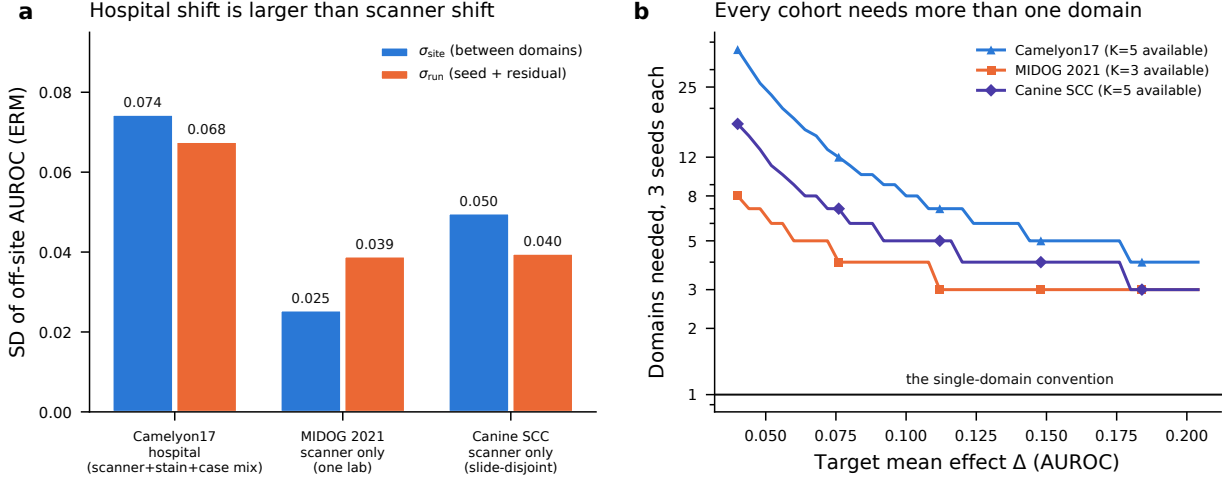


Figure 4: **(a)** ERM variance components per cohort. The between-domain term shrinks as the domain variable narrows from a hospital to a scanner, while the run-to-run term stays put and on MIDOG overtakes it. **(b)** Domains needed for 80% power at three seeds. No cohort is served by one.

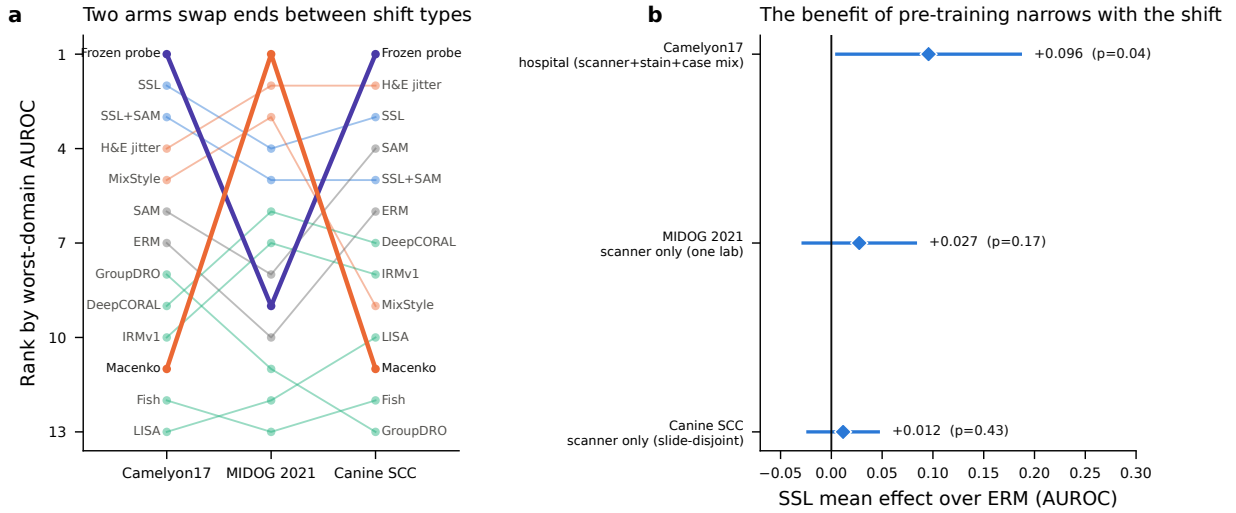


Figure 5: **(a)** Rank by worst-domain AUROC across the three cohorts, for the thirteen arms common to all. Macenko (orange) and the frozen probe (violet), highlighted, swap ends between hospital shift and scanner-only shift. **(b)** SSL mean effect over ERM with a paired 95% interval across domains. The advantage is confined to the cohort whose domains differ by more than the scanner.

How many hospitals is enough. Under the same model, the standard error of a mean-across-sites contrast from a K -hospital, s -seed design is $\sqrt{\sigma_{\text{site}}^2/K + \sigma_{\text{run}}^2/(Ks)}$, where σ_{run} collects the seed and residual terms. Solving for the smallest K at which a paired t -test reaches 80% power gives Table 6 and Figure 6b. Detecting a 0.10 AUROC mean effect needs eleven hospitals at one seed each, or eight at three seeds. Camelyon17 provides five. A single-hospital ablation reporting a mean effect below roughly 0.20 AUROC is therefore under-powered by an order of magnitude. This is also why we prefer the worst-site and across-site-spread summaries: they say something at $K = 5$, whereas the mean does not.

Table 5: Variance components of off-site AUROC, estimated per arm from the 5×3 grid under $y_{ts} = \mu + b_t + e_{s(t)}$. $\text{ICC}_{\text{site}} = \sigma_{\text{site}}^2 / (\sigma_{\text{site}}^2 + \sigma_{\text{seed}}^2 + \sigma_{\text{resid}}^2)$ is the share of variance attributable to hospital identity.

Arm	σ_{site}	σ_{seed}	σ_{resid}	$\text{ICC}_{\text{site}}^*$
ERM	0.074	0.019	0.065	0.55
SAM	0.028	0.056	0.119	0.04
SSL	0.020	0.000	0.027	0.36
SSL+SAM	0.028	0.000	0.012	0.84

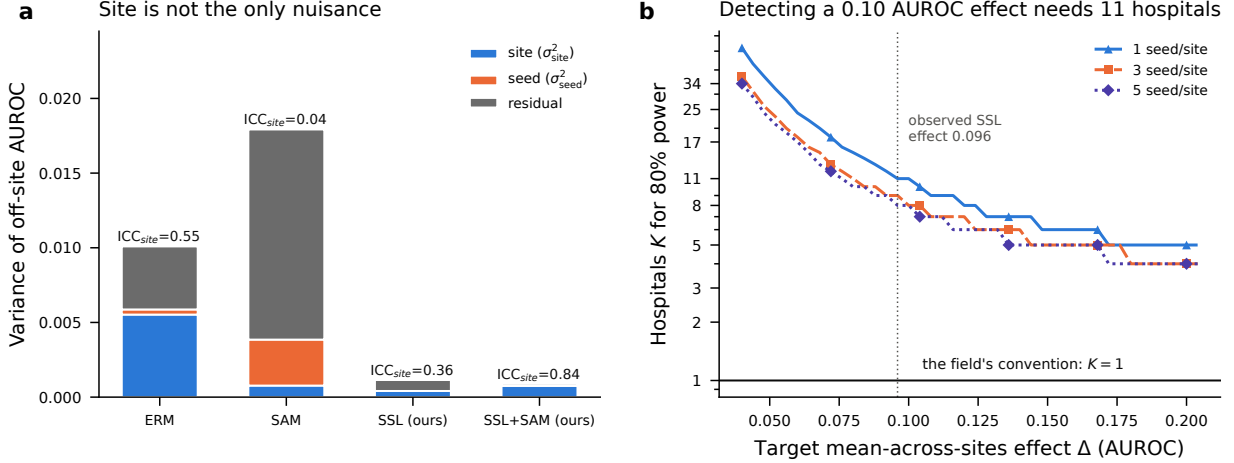


Figure 6: **(a)** Variance of off-site AUROC decomposed into site, seed and residual components. SAM’s variance is almost entirely non-site; SSL+SAM’s is almost entirely site, at one eighth the magnitude. **(b)** Hospitals required for 80% power at $\alpha = 0.05$ as a function of the target mean effect, under the baseline’s variance components.

6 Discussion

Why domain shift type matters more than domain shift magnitude. The most important finding in this paper is not that single-hospital evaluation is unreliable—that is the premise—but that the *kind* of shift between domains governs which method works. Hospital shift bundles three distinct sources: scanner hardware, staining chemistry, and patient/case-mix variation. When all three change together (Camelyon17), the effective intervention is one that builds a more general representation—self-supervised pre-training or aggressive augmentation—because no single normalization step can undo a compound shift. When only the scanner changes (MIDOG, canine cohort), stain normalization removes the dominant signal directly and nothing else is needed. The frozen-probe result sharpens this further: a general-purpose ImageNet representation already encodes enough about tissue structure that it is largely invariant to hospital-level confounds, which explains why it ranks first under hospital shift but drops under scanner-only shift, where the task (mitosis classification) demands fine morphological features that ImageNet representations do not capture. For developers building clinical AI, the implication is that robustness is not a single axis with a single solution. The first step is to characterize the deployment shift—is it a scanner change, a protocol change, or a full institutional change?—and then select the intervention that addresses that specific source.

The ICC as a design tool for robustness evaluation. Current evaluation practice treats site as the only nuisance variable. Our variance decomposition shows this is incomplete: on the supervised baseline, hospital identity accounts for 55% of the variance ($\text{ICC}_{\text{site}} = 0.55$), but on SAM it accounts for only 4%; the rest is seed and residual noise. Without the decomposition, SAM’s across-site SD of 0.074 looks comparable to the baseline’s 0.083, but the ICC reveals that SAM’s spread is instability rather than site sensitivity—a

Table 6: Hospitals K needed to detect a mean-across-sites effect Δ with 80% power at $\alpha = 0.05$ (two-sided), under $\sigma_{\text{site}} = 0.074$ and $\sigma_{\text{run}} = 0.068$ estimated from the baseline arm of this grid.

Δ (AUROC)	0.05	0.075	0.10	0.15	0.20
K , 1 seed/site	34	17	11	6	5
K , 3 seeds/site	25	12	8	5	4
K , 5 seeds/site	23	12	8	5	4

distinction invisible to any summary that does not separate the two sources. The ICC also provides a direct path to statistical power. Our closed-form expression for the standard error of a mean-across-sites contrast yields a concrete design rule: detecting a 0.10 AUROC effect at 80% power requires eight hospitals at three seeds. Camelyon17’s five hospitals fall short, which is why we recommend worst-site and across-site spread as primary summaries: they remain informative at $K = 5$ where the mean does not. We propose that future robustness papers report the ICC alongside their accuracy numbers, so that readers can judge whether the evaluation grid is powered to support the claimed effect.

Implications for clinical AI deployment. For companies and hospitals developing pathology AI for multi-site deployment, our results suggest a three-step workflow. First, *characterize the shift*: determine whether the target sites differ from the training sites in scanner, staining protocol, patient population, or all three. Second, *select the intervention accordingly*: use stain normalization for scanner-only shifts; use self-supervised pre-training or strong colour/stain augmentation for full institutional shifts; invest in stronger backbone representations when possible, since the frozen-probe result shows this is the most effective single lever. Third, *validate domain-completely*: run the model at every available site rather than one, report worst-site performance as the deployment-relevant metric, replicate seeds, and compute the ICC to verify that the evaluation has sufficient power. The protocol costs under one hour of GPU time at compact-model scale—less than a single conventional ablation.

What the grid corrected in our own work. Three findings contradicted what we expected. The hospital effect is not the dominant nuisance: once seeds are replicated, random variation accounts for roughly half the total variance, and on MIDOG it exceeds the hospital effect. A gain-versus-baseline diagnostic we had used to argue that our method targets the hard hospitals is negative by construction (Appendix F): any uniformly better method produces the same pattern with no actual targeting. And the cross-cohort comparison undercut the method ranking we would have published from Camelyon17 alone. All three corrections fell out of running the evaluation completely and repeating it on data with a different shift structure.

Caveats. The most important caveat is model scale. Every trained arm uses a 0.58 M-parameter CNN at 64×64 resolution, which makes the protocol affordable but limits how far the conclusions generalize. The frozen-probe result suggests that larger representations may dissolve the hospital-to-hospital gap entirely; whether the patterns we report survive fine-tuning of a pathology foundation model (Chen et al., 2024) is the most important open question. PatchCamelyon, our out-of-partition test, shares the consortium and scanners with our training data and is therefore not truly independent external validation; the absolute AUROC values (0.71–0.78) confirm that these are comparative instruments, not deployment-ready models. Pre-training only helps under institutional shift: on NCT-CRC-HE (Appendix D), where the shift is a same-institution stain perturbation, every intervened arm underperforms the baseline. Appendix A collects further caveats including re-implementation fidelity, non-independent folds, and unmatched compute budgets.

7 Conclusion

Cross-domain generalization in histopathology is not a single problem with a single solution. By running eleven methods across three cohorts that isolate different sources of shift, we show that the type of domain difference—scanner, staining protocol, or full institutional change—determines which robustness strategy succeeds and which fails. Under hospital shift, self-supervised pre-training compresses the across-site spread

threefold and raises the worst hospital by 0.175 AUROC; under scanner-only shift, stain normalization dominates and pre-training adds nothing. Most domain-generalization objectives hurt the worst domain across all three cohorts. A frozen ImageNet probe outperforms every trained-from-scratch arm under hospital shift, establishing representation quality as a stronger lever than any robustness objective at this scale.

We propose using the intraclass correlation coefficient from a variance decomposition of off-site performance as the field’s standard measure for evaluating robustness claims. This single statistic separates true site sensitivity from seed noise, yields a closed-form power calculation for experimental design, and exposes under-powered evaluations before they produce misleading conclusions.

For the development of clinical pathology AI with practical cross-domain generalization: characterize the deployment shift before choosing an intervention, validate domain-completely at every available site, report worst-site performance as the deployment-relevant metric, replicate seeds, and use the ICC to verify that the evaluation grid supports the claimed effect. The complete protocol costs under one hour of GPU time—there is no computational barrier to doing this correctly.

Broader impact statement

This work uses only de-identified public data releases; no new patient data were collected and no ethics approval was required. Its direct intent is to make cross-hospital robustness claims in medical imaging harder to overstate, which we view as a safety contribution. The models reported here are research instruments trained at reduced resolution with low absolute accuracy and are explicitly not suitable for clinical use; we flag this because released checkpoints in this area are sometimes reused out of context.

Reproducibility statement

Camelyon17-WILDS (Koh et al., 2021) and PatchCamelyon (Veeling et al., 2018) are public under CC0; NCT-CRC-HE and CRC-VAL-HE-7K (Kather et al., 2018; 2019) are public under CC-BY; MIDOG 2021 (Aubreville et al., 2023) and the multi-scanner canine cohort (Wilm et al., 2023) are public on Zenodo under CC-BY. Code, the fold-materialization scripts for all four cohorts, the per-run JSON records for all 388 GPU jobs, the variance-decomposition and power code, and the single script that regenerates every table and figure in this paper from those records are available at <https://github.com/InvenTide-AI/histopath-shift>.

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A Further caveats

Re-implementation fidelity. Our Fish and LISA arms are not the code the original authors ran. They were given no per-arm tuning, they sit inside a 0.58M-parameter five-epoch pipeline rather than the DenseNet-scale setups of the original papers, and in Fish’s case the out-of-distribution validation curve selects epoch 1 in four of five folds (Figure 10). Table 1 is thirteen arms under one fixed protocol, not a ranking of these methods at their best. Our claim about them concerns how much their off-site accuracy moves when the held-out hospital changes, which is a property of the protocol and not of the implementation.

The folds are not independent draws. Each of the five centres appears in three of the five training sets, so the folds share training data and the fold-level intervals we report are approximate and mildly optimistic. The power calculation in §5.6 inherits the same approximation. A cohort with more centres would let the grid be built from disjoint training sets; Camelyon17 does not.

Compute is not matched. The pre-trained arms consume additional unlabelled data and additional optimization steps that the supervised arms do not, and the frozen-probe arm rests on an ImageNet-1K pre-training budget that is not counted anywhere in our totals. A compute-matched control, and an augmentation-only control using the same colour and stain jitter without the contrastive objective, would separate the representation learning from the extra budget. We ship both as scripts but have not run them at the full grid.

B Per-arm, per-hospital results

Figure 7 gives every arm at every hospital. The ordering of arms changes substantially from hospital to hospital, which is the paper’s central practical point in one picture: at centre 0 all thirteen arms fall within 0.106 AUROC of one another, while at centre 2 the same thirteen arms span 0.420.

Figure 8 shows every individual run of the four core arms, making the seed spread visible without summary statistics.

C Full per-arm tables for the two scanner cohorts

Columns match Table 1. Both cohorts were run through the same pipeline, the same arm code and the same aggregator as Camelyon17; the aggregator reproduces the Camelyon17 numbers in Table 1 exactly when pointed at that cohort’s records.

The excess response-shape statistic behaves the same way here as on Camelyon17 and reinforces Appendix F: the arms at the top of each table sit at or near zero excess, meaning they are uniformly better rather than selectively repairing weak domains, while GroupDRO, MixStyle, IRMv1 and RotInv carry large positive excess on one cohort or the other, sending what gain they have to the domains that were already handled.

D Second task: NCT-CRC-HE

To test whether the benefit of pre-training is specific to institutional shift we repeated the four core arms on NCT-CRC-HE (Kather et al., 2018), a colorectal H&E task (tumour epithelium versus all other tissue, tiles resized to 64×64), with 20,000 training and 4,000 in-distribution validation tiles from the colour-normalized release and 10,226 further tiles held unlabelled for pre-training (drawn from the colour-normalized cohort only, so the encoder never sees the perturbed appearance that forms the test condition). Two shift axes are available, neither of which is a change of institution: a stain-appearance perturbation of the training tissue, and the patient-disjoint CRC-VAL-HE-7K release from the same tissue bank (Kather et al., 2019). Two seeds per arm. One protocol difference from §3 is forced by the data: NCT-CRC-HE has no second site, so epochs are selected on in-distribution validation rather than on a held-out validation hospital.

Table 9 is a clean negative result. On the stain-perturbation axis SSL costs 0.089 AUROC relative to the baseline; on the patient-disjoint external cohort all four arms are within 0.017 AUROC of one another, and

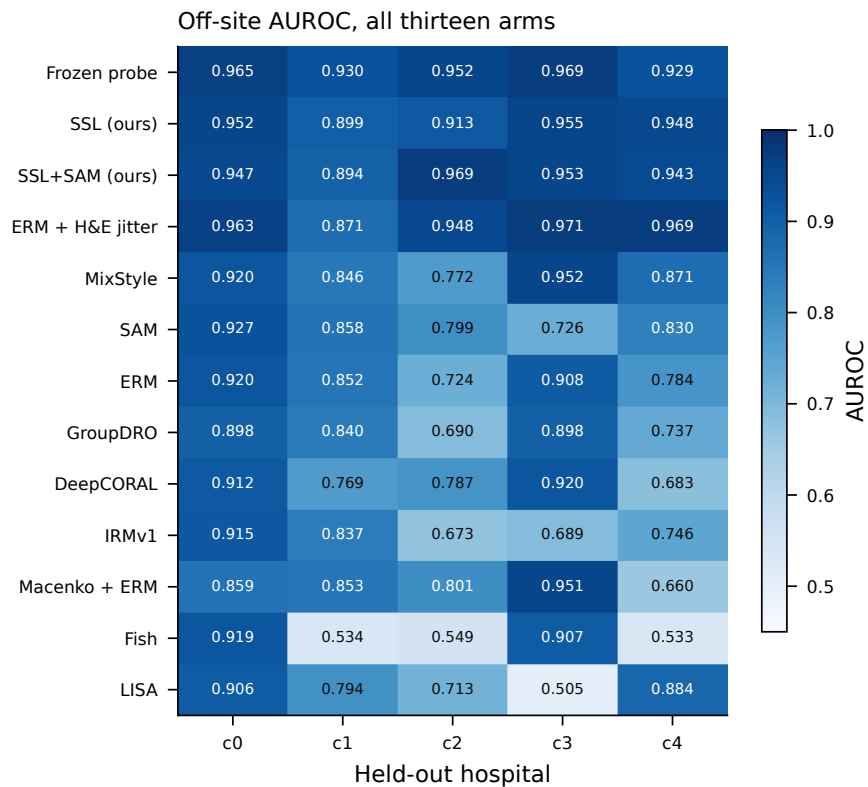


Figure 7: Off-site test AUROC for all thirteen arms at all five held-out hospitals, seed-averaged where multiple seeds were run.

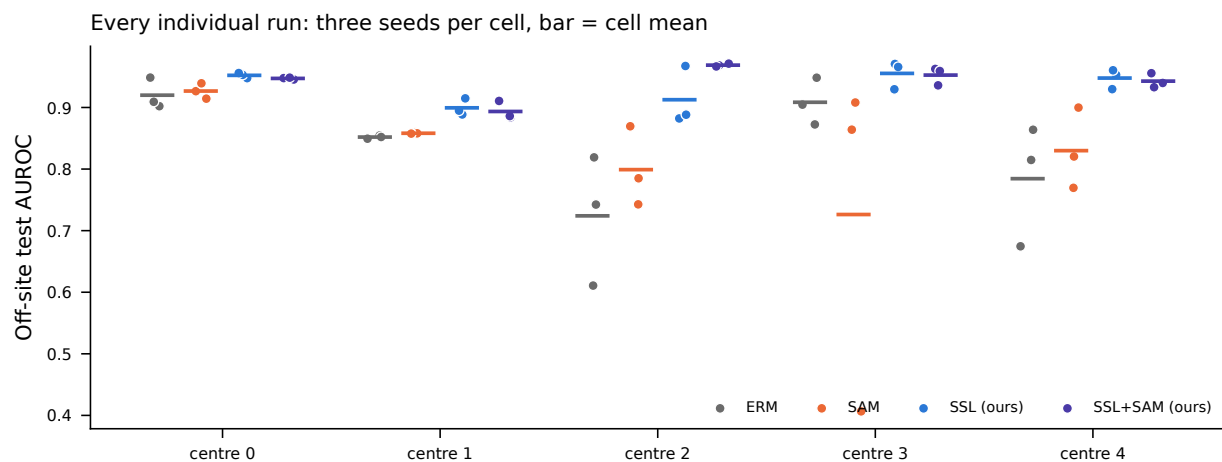


Figure 8: All 60 individual runs of the four core arms. Horizontal bars are cell means. The baseline’s spread at centres 2 and 4, and SAM’s at centre 3, are seed effects rather than site effects.

the baseline is better calibrated than any intervened arm on both axes. We read this as a scope statement on §5.2: pre-training on in-domain unlabelled tissue earns its cost where the shift is a genuine change of institution, and not where it is a same-institution appearance perturbation. Because NCT-CRC-HE is single-

ROC at every held-out hospital (seed 0)

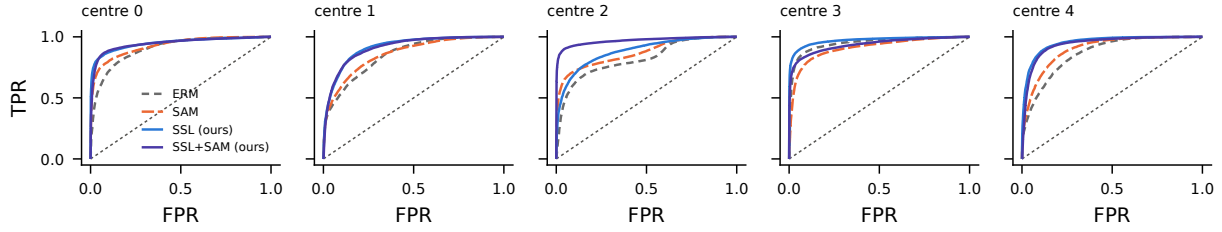


Figure 9: ROC curves at every held-out hospital, seed 0.

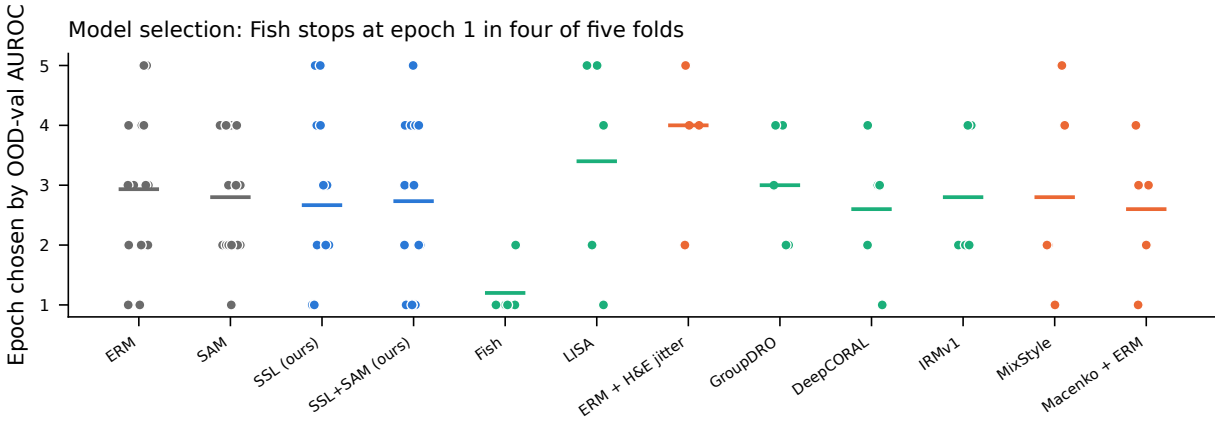


Figure 10: Epoch selected by out-of-distribution validation AUROC, every run. Fish selects epoch 1 in four of five folds, which contributes to its low worst-site score in Table 1 and should be read as a model-selection pathology under our protocol rather than as a property of the method.

institution, it also offers no site structure to replicate over. It is a $K = 1$ design by construction, and Table 6 applies to it.

E A finding of our own that the site-complete grid overturned

We record this because it motivated the study and because it illustrates the protocol’s value on our own work. An earlier single-hospital experiment of ours at Camelyon17 centre 2 reported a synergy of $+0.161$ AUROC between pre-training and sharpness-aware minimization, decomposed as $\delta_{\text{SSL}} = -0.037$, $\delta_{\text{SAM}} = -0.105$ and $I = +0.161$, and interpreted the interaction as a property of the method pair.

Under the site-complete grid that interpretation does not survive. The interaction term has a five-hospital mean of $+0.017$ AUROC (95% CI $[-0.097, +0.132]$, $p = 0.70$) and changes sign across hospitals (Table 2); at centre 2 itself, with three seeds instead of one, it is -0.019 . The original estimate was driven by a single anomalous SAM run: because I is a difference of four arm-level estimates, one bad cell propagates into it undiminished, and SAM is the least stable arm in the study (seed SD 0.084, and 0.278 at centre 3). The reading we now think correct is that $+0.161$ described one hospital and one unlucky run, and the synergy claim is withdrawn. What does survive is this paper’s positive result: the worst-site lift and the variance compression of pre-training itself, both of which are larger and better supported under $K = 5$ with three seeds than anything the $K = 1$ experiment could have established.

Table 7: MIDOG 2021, scanner-complete, $K = 3$. Mitotic figure versus annotated non-mitotic look-alike, all 150 annotated regions. Ordered by worst-scanner AUROC. Note Macenko at the top and the frozen probe in tenth: both are the reverse of their Camelyon17 positions.

Arm	Worst	SD	Mean	Seed SD	ECE	r	r_0	$r - r_0$
Macenko + ERM	0.865	0.013	0.880	—	0.078	-0.93	-0.93	+0.00
TIA-style	0.860	0.022	0.885	—	0.053	-0.75	-0.83	+0.08
ERM + H&E jitter	0.834	0.026	0.863	—	0.058	-0.60	-0.78	+0.18
MixStyle	0.790	0.010	0.799	—	0.265	-1.00	-0.95	-0.04
SSL (ours)	0.789	0.018	0.807	0.022	0.226	-0.85	-0.88	+0.03
SSL+SAM (ours)	0.783	0.016	0.798	0.015	0.277	-0.98	-0.89	-0.08
DeepCORAL	0.776	0.033	0.813	—	0.227	-0.45	-0.71	+0.25
IRMv1	0.775	0.033	0.800	—	0.244	+0.58	-0.70	+1.28
SAM	0.765	0.029	0.784	0.018	0.268	-0.51	-0.75	+0.24
Frozen probe	0.756	0.012	0.768	—	0.075	-0.93	-0.94	+0.00
ERM baseline	0.754	0.033	0.779	0.034	0.240	—	—	—
GroupDRO	0.750	0.034	0.783	—	0.293	-0.06	-0.70	+0.63
RotInv	0.732	0.059	0.788	—	0.275	+0.88	-0.49	+1.36
LISA	0.680	0.061	0.750	—	0.309	+0.26	-0.48	+0.73
Fish	0.578	0.054	0.637	—	0.173	-0.52	-0.52	+0.01

Table 8: Canine multi-scanner SCC, scanner-complete, $K = 5$, scale-normalized and slide-disjoint. Tumour versus non-tumour tissue. This is the one cohort where the Camelyon17 protocol applies without adaptation.

Arm	Worst	SD	Mean	Seed SD	ECE	r	r_0	$r - r_0$
Frozen probe	0.792	0.038	0.845	—	0.094	-0.70	-0.81	+0.11
TIA-style	0.773	0.034	0.821	—	0.055	-0.80	-0.84	+0.04
ERM + H&E jitter	0.756	0.029	0.777	—	0.043	-0.86	-0.88	+0.02
SSL (ours)	0.703	0.038	0.744	0.037	0.143	-0.72	-0.81	+0.09
SAM	0.702	0.060	0.744	0.015	0.143	-0.06	-0.66	+0.61
SSL+SAM (ours)	0.697	0.047	0.752	0.032	0.114	-0.53	-0.76	+0.22
ERM baseline	0.694	0.054	0.733	0.034	0.198	—	—	—
DeepCORAL	0.678	0.054	0.734	—	0.146	-0.24	-0.70	+0.46
IRMv1	0.673	0.061	0.755	—	0.141	-0.39	-0.66	+0.27
MixStyle	0.669	0.082	0.742	—	0.162	+0.67	-0.55	+1.22
LISA	0.635	0.071	0.740	—	0.251	-0.36	-0.60	+0.25
Macenko + ERM	0.615	0.098	0.733	—	0.139	+0.33	-0.48	+0.81
Fish	0.603	0.089	0.715	—	0.156	-0.23	-0.51	+0.28
RotInv	0.597	0.081	0.701	—	0.188	+0.02	-0.55	+0.57
GroupDRO	0.518	0.119	0.697	—	0.155	+0.36	-0.41	+0.77

F The response-shape diagnostic

An intervention for site robustness ought to help most where the baseline fails and little where it already succeeds. The natural way to check is the correlation r between an arm’s per-hospital gain and the baseline’s per-hospital AUROC, and a large negative r looks like evidence of exactly that targeting. We used this statistic, and it is misleading.

Because the gain is $M - E$ and is correlated against E itself, the statistic is non-zero even when an arm’s per-hospital performance is statistically independent of the baseline’s. For independent M and E with across-site standard deviations σ_M, σ_E ,

$$\text{Cov}(M - E, E) = -\sigma_E^2, \quad r_0 = \frac{-\sigma_E^2}{\sigma_E \sqrt{\sigma_M^2 + \sigma_E^2}} = \frac{-\sigma_E}{\sqrt{\sigma_M^2 + \sigma_E^2}}.$$

Table 9: NCT-CRC-HE, mean over two seeds. Neither shift axis is an institution change, and on both the intervened arms fail to beat the baseline.

Arm	In-distribution val		Stain perturbation		CRC-VAL-HE-7K	
	AUROC	ECE	AUROC	ECE	AUROC	ECE
ERM	0.972	0.010	0.791	0.156	0.940	0.027
SAM	0.966	0.017	0.726	0.206	0.944	0.037
SSL	0.969	0.011	0.702	0.235	0.927	0.032
SSL+SAM	0.963	0.017	0.724	0.232	0.936	0.036

An arm that is merely *flat* across hospitals ($\sigma_M \rightarrow 0$) therefore has $r_0 \rightarrow -1$ with no targeting whatever. Our own arms are exactly in that regime: SSL has $r = -0.95$ against $r_0 = -0.96$, and SSL+SAM $r = -0.96$ against $r_0 = -0.95$ (Table 1, Figure 11). The excess is zero to two decimal places. Our pre-trained arms do not preferentially repair the hard hospitals; they are uniformly better, and the apparent targeting was an artifact of the statistic.

Removing the null makes the statistic informative again, and what it finds is the reverse of the naive reading. GroupDRO sits +1.37 above its null and Fish +0.97: these arms concentrate what gain they have at the hospitals where the baseline is *already* strong, widening the cross-hospital gap they are meant to close. DeepCORAL (+0.51), Macenko (+0.46), IRMv1 (+0.35) and MixStyle (+0.21) show the same misallocation more mildly. We recommend reporting $r - r_0$ with r_0 from Equation 1, and reporting it alongside rather than instead of the worst-site value.

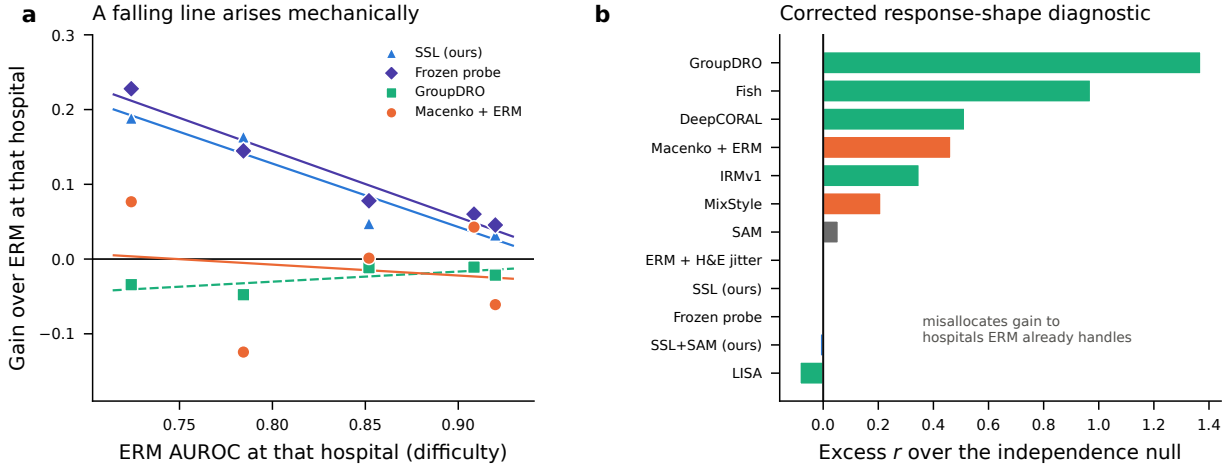


Figure 11: **(a)** Per-hospital gain against per-hospital baseline difficulty for one arm from each family. The pre-trained and frozen arms show steep negative slopes; so would any arm with low across-site variance, by Equation 1. **(b)** Excess correlation over the independence null. Positive excess means the arm sends its gain to hospitals the baseline already handles.